Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_04

1. Summary of Key Deviations

The analysis of Run_04 shows a consistent and moderate loss of flow throughout the system, with a particularly unusual signature in the bypass loop.

- **Flow_Bypass (Moderate):** The most significant anomaly is a large **negative offset** on the bypass flow, which reads **-20% to -25%** lower than healthy. This is the inverse of what is seen in blockage scenarios and is a key diagnostic clue.
- **Flow_Pump (Moderate):** The pump flow shows a corresponding moderate **negative offset** of approximately **-9.5%**. This indicates a system-wide loss of flow originating at or before the pump.
- **p_Combustion (Minor):** A minor **negative offset** of around **-6%** is observed in the combustion pressure.
- **Flow_Engine1-4 (Minor):** Intriguingly, the engine flows show a slight **positive offset** of **+3.4%** during the high-throttle phase (s5), which is counter-intuitive given the overall loss of pump flow.
- **Other Signals (Nominal):** All other pressure and flow signals show only negligible deviations.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Bypass Valve**	**Stuck Significantly Open**	This is the most compelling hypothesis. If the bypass valve is stuck open, it diverts a large portion of the pump's output directly back to the inlet. This means the `Flow_Bypass` sensor (which is in the *controlled* bypass line) would read near zero, and the `Flow_Pump` sensor would also read low because much of its output is immediately lost to the leak before it can be properly measured or used. This perfectly explains the large negative offsets on both `Flow_Bypass` and `Flow_Pump`.

2	**Main Fuel Pump**	**Moderate Internal Leakage / Degradation**	A degraded pump would explain the `Flow_Pump` drop of -9.5%. However, it doesn't as elegantly explain why the `Flow_Bypass` reading is so disproportionately low (-25%). A stuck-open valve provides a better explanation for the specific relationship between these two signals.
3	**Boost-Pressurised Tank / Inlet Line**	**Moderate Blockage**	A moderate blockage before the pump would starve it of fuel, causing a drop in `Flow_Pump` and `Flow_Bypass`. This is a plausible alternative for the physical fault, but again, doesn't explain the particularly severe drop in the bypass reading as well as a stuck valve does.
4	**Flow_Bypass Sensor**	**Sensor Dropout / Negative Offset**	It's possible the `Flow_Bypass` sensor has failed with a large negative offset. However, the `Flow_Pump` reading is *also* low, as are other system pressures. This indicates a real physical loss of flow, making a sensor-only fault unlikely.
5	**Fuel Metering Unit (FMU)**	**Control System Error**	It's possible a control error is causing the FMU to command the bypass valve to stay open while also trying to compensate by adjusting engine flow, leading to the unusual readings. However, a mechanical failure of the valve is a more direct explanation.

3. Most Probable Hypothesis

The most probable root cause is that the **Bypass Valve is Stuck Significantly Open.**

Reasoning: This single mechanical failure provides the most complete explanation for the unique data signature. When the bypass valve sticks open, a large portion of the pump's output is immediately "leaked" back to the pump inlet. This has two key effects seen in the data: 1) The overall flow available to the rest of the system drops, causing the `Flow_Pump` reading to be low (-9.5%). 2) The flow through the *measured and controlled* part of the bypass circuit, where the `Flow_Bypass` sensor is, also drops dramatically (-25%) because most of the diverted flow is going through the unintended leak path. This one fault elegantly explains both major negative flow deviations, which other hypotheses struggle to reconcile.